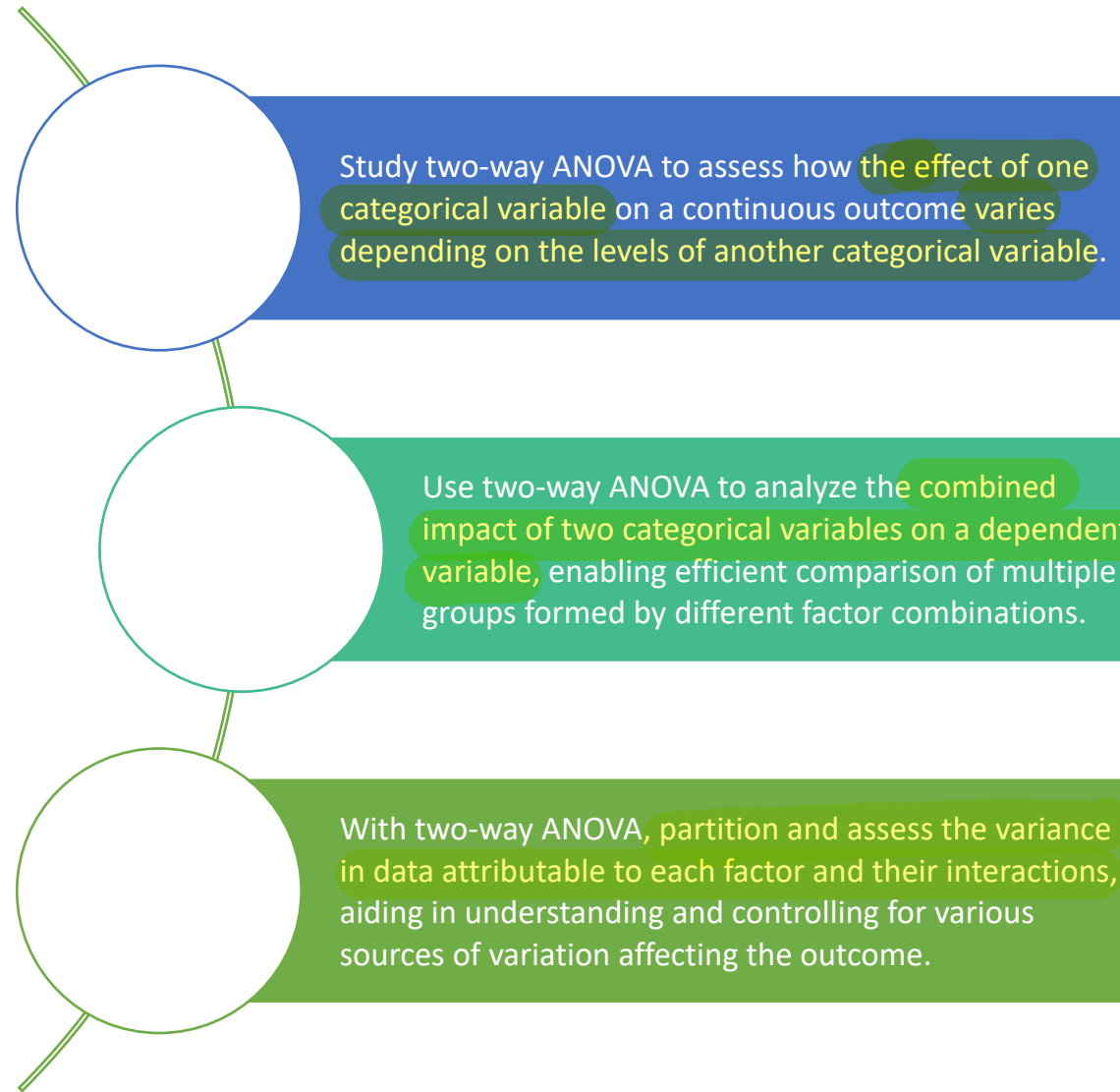




Stat 3113: Lesson 28

Two-Way ANOVA

Motivations





Two-Factor Experiments

- In **one-factor** experiments, the purpose is to determine whether **varying the level of a single factor** affects the response.
- Many experiments involve varying **several factors**, each of which may affect the response.
- In this section, we will discuss the case in which there are two factors. The experiments are called **two-factor experiments**.
- This can be observational studies also.

Example 1:

- There are two factors, the catalyst and reagent.
- Four runs (replicates) of the process were made for each treatment combination.
- The catalyst is called the row factor since its values varies from row to row in the table.
- The reagent is called the column factor.

TABLE 9.2 Yields for runs of a chemical process with various combinations of reagent and catalyst

Catalyst	Reagent											
	1				2				3			
A	86.8	82.4	86.7	83.5	93.4	85.2	94.8	83.1	77.9	89.6	89.9	83.7
B	71.9	72.1	80.0	77.4	74.5	87.1	71.9	84.1	87.5	82.7	78.3	90.1
C	65.5	72.4	76.6	66.7	66.7	77.1	76.7	86.1	72.7	77.8	83.5	78.8
D	63.9	70.4	77.2	81.2	73.7	81.6	84.2	84.9	79.8	75.7	80.5	72.9

-
- In general, there are I levels of the row factor and J levels of the column factor. (In our example, $I = 4$ and $J = 3$.) There are therefore $I * J$ (12) different combinations of the two factors.
 - The units assigned to a given treatment are called replicates. This is just the sample size within each treatment group. When the number of replicates is the same for each treatment, we will denote this number by K . Thus in our example, $K = 4$.
 - When observations are taken on every possible treatment, the design is called a **complete design** or a **full factorial design**.

Notes

- Incomplete designs, in which there are no data for one or more treatments, can be difficult to interpret.
- When possible, complete designs should be used.
- When the number of replicates is the same for each treatment, the design is said to be **balanced.**
- With two-factor experiments, unbalanced designs are much more difficult to analyze than balanced designs.
- The factors may be fixed or random.
- The discussion on the following slides applies only to balanced designs with fixed effects.

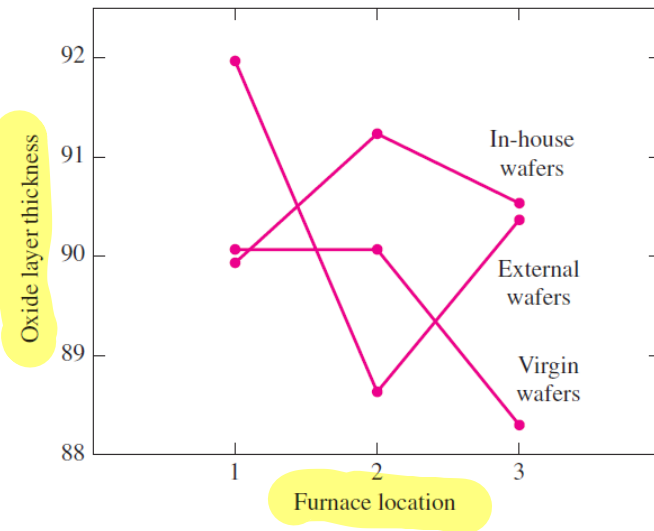
Main Effects in Two-Way ANOVA

- Let's call the two factors *Factor A* and *Factor B*.
- Both Factor A and Factor B are said to be **main effects**. This describes the **impact each individual factor has on the response variable**. So, we have the Main Effect of Factor A and the Main Effect of Factor B.
- These main effects look at each factor's influence on the response variable on its own **(without considering the other factor)**. What effect does changing the levels of this one factor have on the response variable?
- In our example, we have the **main effect of catalyst** AND the **main effect of reagent**.

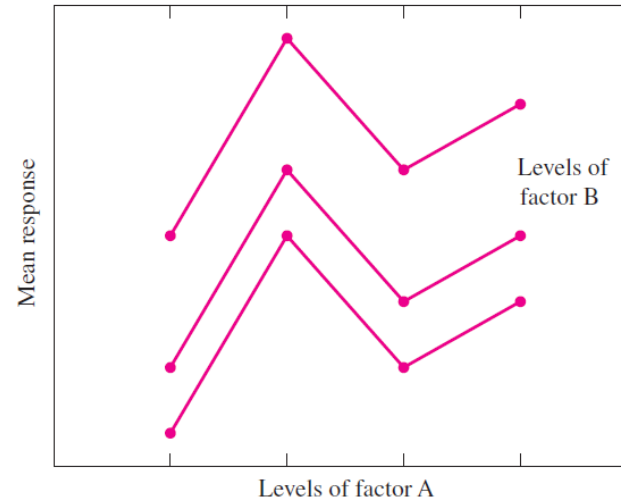
Interaction Effect in Two-Way ANOVA

- The interaction of two factors is called a **two-way interaction effect**.
- The interaction effect results from combining the levels of factor A with the levels of factor B.
- We look at the influence of changing the levels of factor A under one level of factor B. Then, we see if this effect, this pattern, for factor A is different when you look at the other level of factor B.
- A two-way interaction effect is present when the relationship between one factor and the response variable changes as the levels of the other factor change.

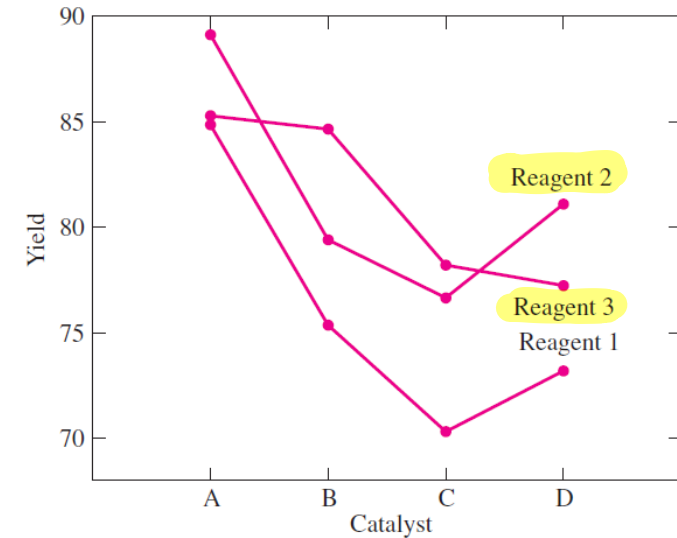
Interaction plot



The lines are far from parallel, indicating substantial interaction between the factors.



Interaction plot for hypothetical data with interaction estimates equal to 0. The line segments are parallel.



Interaction plot for yield data.

The more nonparallel the lines are, the greater the strength of the interaction.

Hypotheses for Two-Way ANOVA

1.

- **First**, test for the presence of interaction effect,

H_0 : *no interaction between two factors*

H_A : *there is an interaction between the two factors*

Hypotheses for the main effects are tested **only if** there is not evidence of a statistically significant interaction effect **i.e. we failed to reject H_0 for the first hypothesis test.**

- **DO NOT** test the main effects **if interaction is present**
- Tests for the main effects have no real meaning if interaction is present.

Hypotheses for Two-Way ANOVA (cont.)

If the interaction effect does NOT exist, then continue to test the main effects! Otherwise, stop any further testing!

Test the main effect following the following procedures:

2. Test for main effects of the row factor (A)

H_0 : mean of response is same for all levels of Factor A

H_A : mean of response is different for at least one level of Factor A

3. Test for main effects of the column factor (B)

H_0 : mean of response is same for all levels of Factor B

H_A : mean of response is different for at least one level of Factor B

Two-Way ANOVA Table

Source of Variation	d.f.	SS	MS	F	p-value
Treatment Combination	$IJ - 1$	SST_r	MST_r	$F_{Tr}^* = MST_r / MSE$	p_{Tr}
Row factor (A)	$I - 1$	SSA	MSA	$F_A^* = MSA / MSE$	p_A
Column factor (B)	$J - 1$	SSB	MSB	$F_B^* = MSB / MSE$	p_B
Interaction (A*B)	$(I - 1)(J - 1)$	$SSAB$	$MSAB$	$F_{AB}^* = MSAB / MSE$	p_{AB}
Error	$N - IJ$	SSE	MSE	-----	
Total	$N - 1$	SST	-----	-----	

Steps for Analysis of Two-Way Experiments

For a set significance level α

1. Check (a) normality assumption and (b) equal variance assumption

2. Test for significance of the interaction:

$pvalue \leq \alpha \rightarrow \text{reject } H_0 \rightarrow \text{interaction significant} \rightarrow \text{STOP}$

$pvalue > \alpha \rightarrow \text{fail to reject } H_0 \rightarrow$
 $\text{interaction not significant} \rightarrow$
 $\text{continue with step 3 and 4}$

3. Test for Factor A effect:

$pvalue \leq \alpha \rightarrow \text{reject } H_0 \rightarrow \text{factor A is significant}$

$pvalue > \alpha \rightarrow \text{fail to reject } H_0 \rightarrow \text{factor A is not significant}$

4. Test for Factor B effect:

$pvalue \leq \alpha \rightarrow \text{reject } H_0 \rightarrow \text{factor B is significant}$

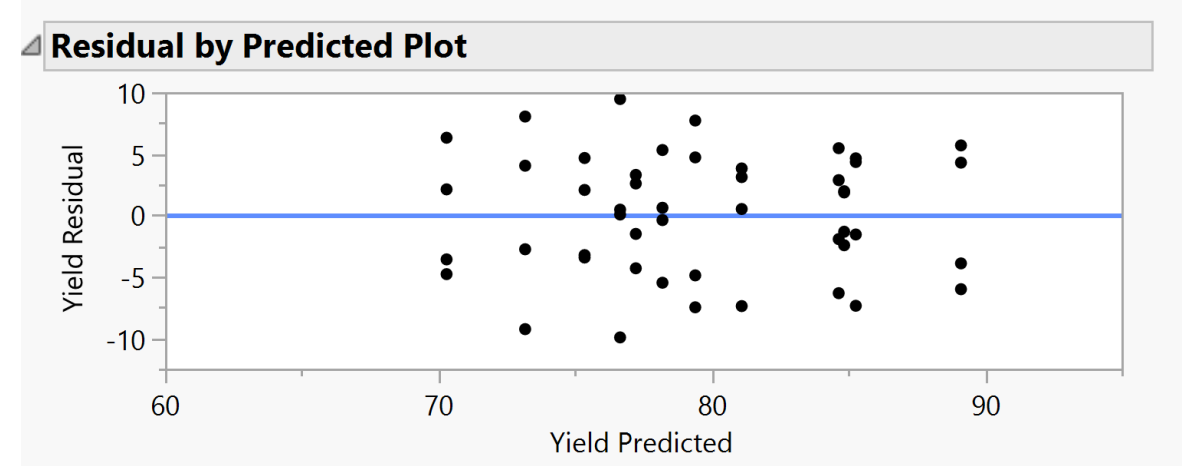
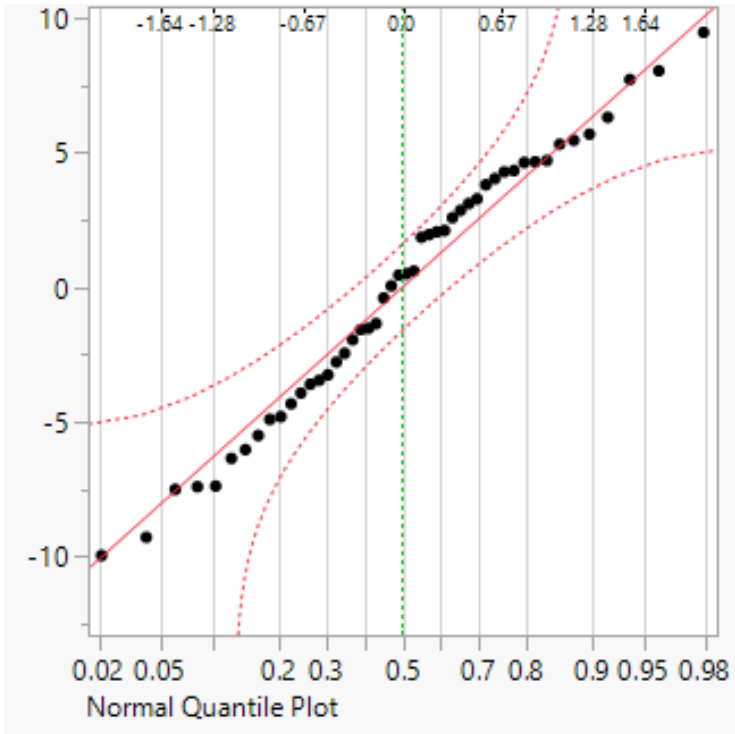
$pvalue > \alpha \rightarrow \text{fail to reject } H_0 \rightarrow \text{factor B is not significant}$

Checking Assumptions

Errors are:

- Normal
 - Normal probability Plot
- Constant Variance
 - Residual vs. Fitted
 - Residual vs. Factor Levels
- Independence
 - Residual vs. Order (if appropriate)





Example: Yield – check assumptions

- Constant Variance and Normality appear to be met.

Example: Yields

Effect Tests					
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Catalyst	3	3	877.56333	9.3579	0.0001*
Reagent	2	2	327.14042	5.2327	0.0101*
Catalyst*Reagent	6	6	156.98292	0.8370	0.5496

→ $p\text{-value} > \alpha \Rightarrow \text{Fail to Reject } H_0$.

Step 1: Test for significance of the interaction.

yes, test for main effects.

Do we test the main effects?

Example: Yields

Effect Tests					
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Catalyst	3	3	877.56333	9.3579	0.0001*
Reagent	2	2	327.14042	5.2327	0.0101*
Catalyst*Reagent	6	6	156.98292	0.8370	0.5496

- Step 2: Test catalyst main effect

→ Significant!

Example: Yields

Effect Tests					
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Catalyst	3	3	877.56333	9.3579	0.0001*
Reagent	2	2	327.14042	5.2327	0.0101*
Catalyst*Reagent	6	6	156.98292	0.8370	0.5496

- Step 3: Test reagent main effect

→ significant!

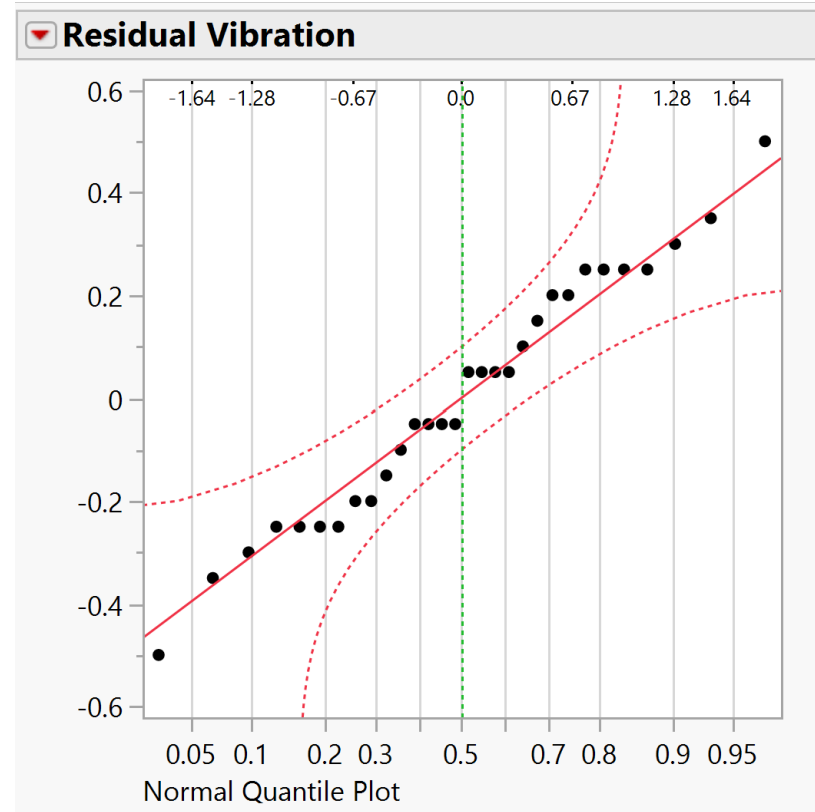
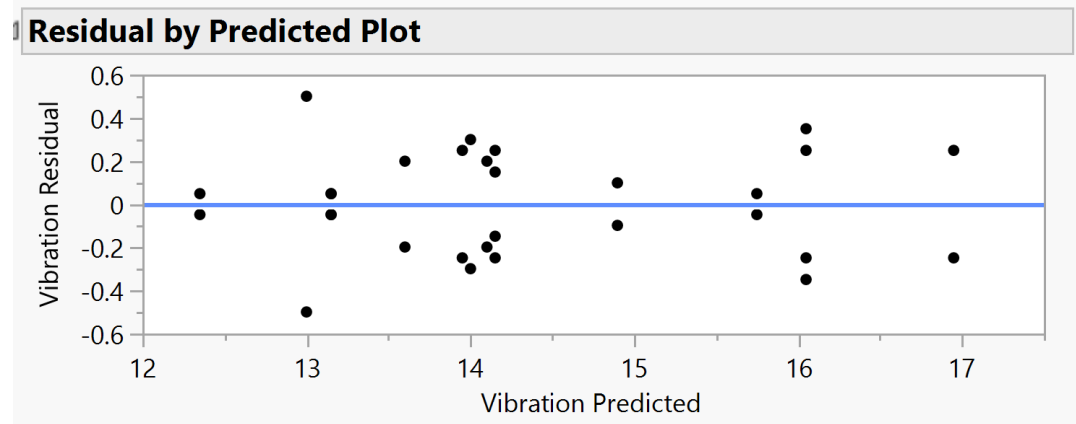
Example: Yield

- There is not a significant interaction between catalyst level and reagent level (p-value=0.5496).
- There is a significant catalyst effect on mean yield (p-value 0.0001).
- There is a significant reagent effect on mean yield (p-value 0.0101).
- We can run multiple comparison tests on both main effects, but that is beyond the scope of our course.

Example:

A process engineer has identified two potential causes of electric motor vibration, the material used for the motor casing (factor *A*) and the supply source of bearings used in the motor (factor *B*). The accompanying data on the amount of vibration (microns) resulted from an experiment in which motors with casings made of steel, aluminum, and plastic were constructed using bearings supplied by five randomly selected sources.

		Supply Source									
		1		2		3		4		5	
Material	Steel	13.1	13.2	16.3	15.8	13.7	14.3	15.7	15.8	13.5	12.5
	Aluminum	15.0	14.8	15.7	16.4	13.9	14.3	13.7	14.2	13.4	13.8
	Plastic	14.0	14.3	17.2	16.7	12.4	12.3	14.4	13.9	13.2	13.1



Example: Check conditions

Constant Variance and Normality appear to be met!

Example: Test Effects

Effect Tests					
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Material	2	2	0.704667	3.1647	0.0713
Supply Source	4	4	36.674667	82.3533	<.0001*
Material*Supply Source	8	8	11.605333	13.0299	<.0001*

→ significant!

Step 1: Test for significance of the interaction.

No, because interaction is significant!

Do we test the main effects?



Reading

- CHAPTER 9 – SECTIONS 9.3

Next Topic



2^3 Factorial Experiment - Analysis of Data



Chapter 9 – Sections 9.5